شهادة إتمام تدريب

CERTIFICATE OF COMPLETION

تبارك الأكاديمية السعودية الرقمية لـ

هشام سعدان السعدان

Hesham Alsadan

إتمام **معسكر تدريبي في تطوير البرمجيات باستخدام بايثون** بنجاح بعد رحلة استمرت لمدة **7 أسابيع** في الفترة من **23 ذو القعدة 1447** حتى **10 محرم 1448**

On successfully completing **Software Development using Python Bootcamp** in a journey that lasted **7 weeks** From **10 May 2026** to **25 June 2026**

تم إصدارها بتاريخ **10 محرم 1448**

Issue date **25 June 2026**

WISHING YOU A CONTINUOUS SUCCESS

مع تمنياتنا بدوام التوفيق والنجاح

Amal A. Aljuhaiman

الرئيس التنفيذي للأكاديمية السعودية الرقمية

امل الجهيمان

Bootcamp Plan

WEEK 1

Pre-bootcamp

- Welcome session (meet and greet)
- Bootcamp overview and introduction
- Self-paced onboarding guide
- Learning support

WEEK 2

Web Fundamentals

- HTML document structure, semantic layout, forms
- CSS selectors, Flexbox/Grid, responsive design, animations
- JavaScript basics, DOM, async programming, API calls
- Git fundamentals
- AI-generated UI with Lovable and VS Code Copilot

WEEK 3

React & Frontend Engineering

- React components, JSX, props, and rendering
- State with useState, lifecycle with useEffect
- Fetch API, async state management, error handling
- Forms, validation, Context API, routing with react-router
- Custom hooks, debugging, and monitoring

WEEK 4

Backend Engineering with FastAPI

- Python review: syntax, typing, async, virtual environments
- FastAPI project structure, routes, Pydantic models
- Dependency injection, path/query parameters
- JWT authentication, password hashing, protected routes
- React + FastAPI integration, CORS, error handling

WEEK 5

Databases & Full-Stack Integration

- SQL basics, normalization, SELECT/JOIN/AGGREGATE
- SQLite setup, migrations, and transactions
- ORM with SQLModel/SQLAlchemy: models and relationships
- FastAPI + DB integration, sessions, dependency injection
- Real client project kickoff

WEEK 6

Applied AI & LLM Integration

- OpenAI API: authentication, rate limits, prompt engineering
- PDF parsing, chunking, and backend LLM endpoint integration
- Concept-only: LLM overview, RAG, LangChain, vector databases, embeddings
- Concept-only: Docker fundamentals

WEEK 7

Capstone Completion & Demo Day

- Project Wrap-up and Final Client Project Presentation